WORKED SOLUTION

STEP III 1994, Question 1

Pure • Reduction Formulae

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s3-q1

The key idea

For the first integral, write $\operatorname{sech} t$ as $\dfrac{\cosh t}{\cosh^2 t} = \dfrac{\cosh t}{1+\sinh^2 t}$: the numerator is the derivative of $\sinh t$, so this is an arctan in disguise. For the reduction formula, the natural move is to split off a factor $\operatorname{sech}^2 t$, which is the derivative of $\tanh t$, and integrate by parts. The identity $\tanh^2 t = 1 - \operatorname{sech}^2 t$ then brings I_n back on the right-hand side, and we collect it.

Solution

The integral of $\operatorname{sech} t$

Multiplying top and bottom by $\cosh t$ and using $\cosh^2 t = 1 + \sinh^2 t$,

$$\int_0^x \operatorname{sech} t \, \mathrm{d}t = \int_0^x \frac{\cosh t}{1+\sinh^2 t} \, \mathrm{d}t.$$

With $s = \sinh t$, $\mathrm{d}s = \cosh t \, \mathrm{d}t$, and the limits become 0 and $\sinh x$, so

$$\int_0^x \operatorname{sech} t \, \mathrm{d}t = \int_0^{\sinh x} \frac{\mathrm{d}s}{1+s^2} = \arctan(\sinh x).$$

An equally good route is $\operatorname{sech} t = \dfrac{2\mathrm{e}^t}{\mathrm{e}^{2t}+1}$ with $u = \mathrm{e}^t$, which gives the answer in the form $2\arctan(\mathrm{e}^x) - \frac{\pi}{2}$. The two agree: both vanish at $x = 0$ and have derivative $\operatorname{sech} x$. (This function of x is called the *Gudermannian*.)

The reduction formula

Take $n \geq 2$ and write $\operatorname{sech}^n t = \operatorname{sech}^{n-2} t \cdot \operatorname{sech}^2 t$. Integrate by parts with

$$u = \operatorname{sech}^{n-2} t, \qquad \frac{\mathrm{d}v}{\mathrm{d}t} = \operatorname{sech}^2 t,$$

so that $v = \tanh t$ and, by the chain rule (using $\frac{\mathrm{d}}{\mathrm{d}t}\operatorname{sech} t = -\operatorname{sech} t \tanh t$),

$$\frac{\mathrm{d}u}{\mathrm{d}t} = (n-2)\operatorname{sech}^{n-3} t \cdot (-\operatorname{sech} t \tanh t) = -(n-2)\operatorname{sech}^{n-2} t \tanh t.$$

Then

$$\begin{aligned} I_n &= \left[\operatorname{sech}^{n-2} t \tanh t\right]_0^x + (n-2)\int_0^x \operatorname{sech}^{n-2} t \tanh^2 t \,\mathrm{d}t \\ &= \operatorname{sech}^{n-2} x \tanh x + (n-2)\int_0^x \operatorname{sech}^{n-2} t\,(1 - \operatorname{sech}^2 t)\,\mathrm{d}t \\ &= \operatorname{sech}^{n-2} x \tanh x + (n-2) I_{n-2} - (n-2) I_n, \end{aligned}$$

where the boundary term at 0 vanishes because $\tanh 0 = 0$.

Ask yourself We seem to have gone in a circle: I_n has reappeared on the right. Is that a problem?

Not at all; it is exactly what we want. The I_n on the right has coefficient $-(n-2)$, not $+1$, so moving it to the left gives $(n-1)I_n$ and we can divide by $n-1 \neq 0$. The reduction formula is

$$\boxed{(n-1)I_n = \operatorname{sech}^{n-2} x \tanh x + (n-2)\, I_{n-2} \qquad (n \geq 2).}$$

Finding I_5

The odd chain starts from $I_1 = \arctan(\sinh x)$. Putting $n = 3$:

$$2I_3 = \operatorname{sech} x \tanh x + I_1, \qquad \text{so} \qquad I_3 = \tfrac{1}{2}\operatorname{sech} x \tanh x + \tfrac{1}{2}\arctan(\sinh x).$$

Putting $n = 5$:

$$4I_5 = \operatorname{sech}^3 x \tanh x + 3I_3,$$

so

$$I_5 = \tfrac{1}{4}\operatorname{sech}^3 x \tanh x + \tfrac{3}{8}\operatorname{sech} x \tanh x + \tfrac{3}{8}\arctan(\sinh x).$$

Finding I_6

The even chain starts from $I_0 = \int_0^x 1\,\mathrm{d}t = x$, or more directly from $I_2 = \int_0^x \operatorname{sech}^2 t\,\mathrm{d}t = \tanh x$ (the formula with $n = 2$ confirms this: $I_2 = \tanh x + 0$). Putting $n = 4$:

$$3I_4 = \operatorname{sech}^2 x \tanh x + 2I_2, \qquad \text{so} \qquad I_4 = \tfrac{1}{3}\operatorname{sech}^2 x \tanh x + \tfrac{2}{3}\tanh x.$$

Putting $n = 6$:

$$5I_6 = \operatorname{sech}^4 x \tanh x + 4I_4,$$

so

$$I_6 = \tfrac{1}{5}\operatorname{sech}^4 x \tanh x + \tfrac{4}{15}\operatorname{sech}^2 x \tanh x + \tfrac{8}{15}\tanh x.$$

The "otherwise" for even powers

Even powers have a neater direct route. With $u = \tanh t$, $\mathrm{d}u = \operatorname{sech}^2 t\,\mathrm{d}t$ and $\operatorname{sech}^2 t = 1 - u^2$, so

$$I_6 = \int_0^x (\operatorname{sech}^2 t)^2 \operatorname{sech}^2 t\,\mathrm{d}t = \int_0^{\tanh x} (1-u^2)^2\,\mathrm{d}u = \tanh x - \tfrac{2}{3}\tanh^3 x + \tfrac{1}{5}\tanh^5 x.$$

Replacing $\operatorname{sech}^2 x$ by $1 - \tanh^2 x$ in our earlier answer and expanding gives exactly this, so the two forms agree.

Checks

- As $x \to \infty$, $\tanh x \to 1$, $\operatorname{sech} x \to 0$ and $\arctan(\sinh x) \to \frac{\pi}{2}$, so $I_5 \to \frac{3\pi}{16} \approx 0.58905$ and $I_6 \to \frac{8}{15}$. Simpson's rule on $[0, 30]$ gives 0.589049 and 0.533333.
- At $x = 1$, Simpson's rule gives $I_5 \approx 0.561566$ and $I_6 \approx 0.518343$, matching the formulae; we also checked the reduction formula for $2 \le n \le 8$ at 200 random values of x.
- Both I_5 and I_6 are odd functions of x, as they must be, since sech is even.

Common slips

- Getting the sign of $\frac{\mathrm{d}}{\mathrm{d}t}\operatorname{sech} t$ wrong: it is $-\operatorname{sech} t \tanh t$, and this minus sign is what makes the boundary term positive.
- Dividing by n instead of $n - 1$ after collecting the two copies of I_n.
- Leaving the answer as an indefinite integral: with limits 0 and x there is no arbitrary constant, and every term should vanish at $x = 0$.

Beyond STEP

Reduction formulae like this one let you evaluate a whole family of integrals by recursion, and the same idea underlies Wallis's formula for $\int_0^{\pi/2} \sin^n t\,\mathrm{d}t$, which leads to a famous infinite product for π. The limits $\int_0^\infty \operatorname{sech}^n t\,\mathrm{d}t$ found here are values of the Beta function, a central object in first-year analysis and probability. Siklos, *Advanced Problems in Mathematics*, has more reduction-formula practice, and Abbott, *Understanding Analysis*, makes the improper integrals rigorous.

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